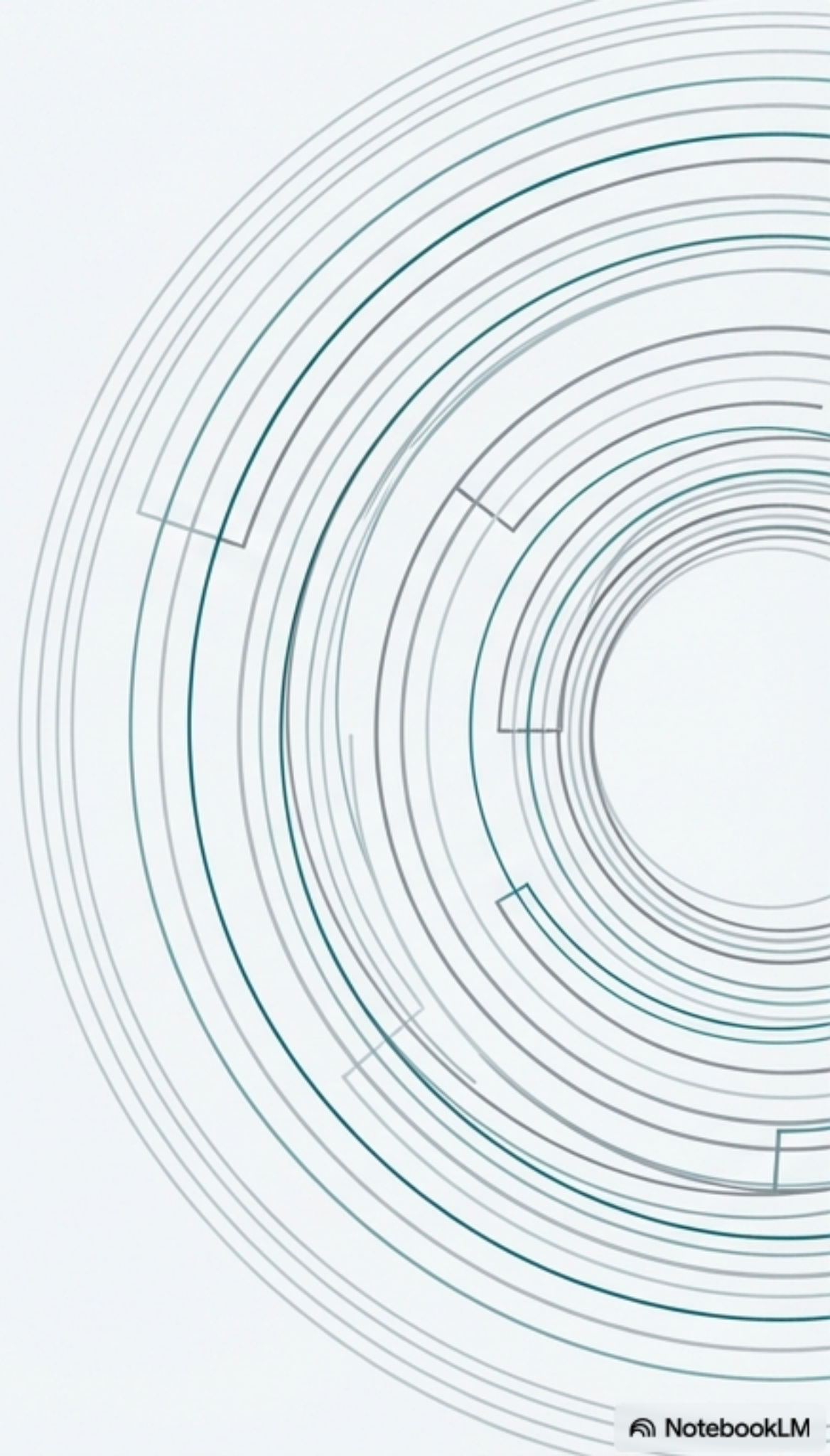


High-Sensitivity Glaucoma Screening via Deep Learning Segmentation of the Optic Nerve Head

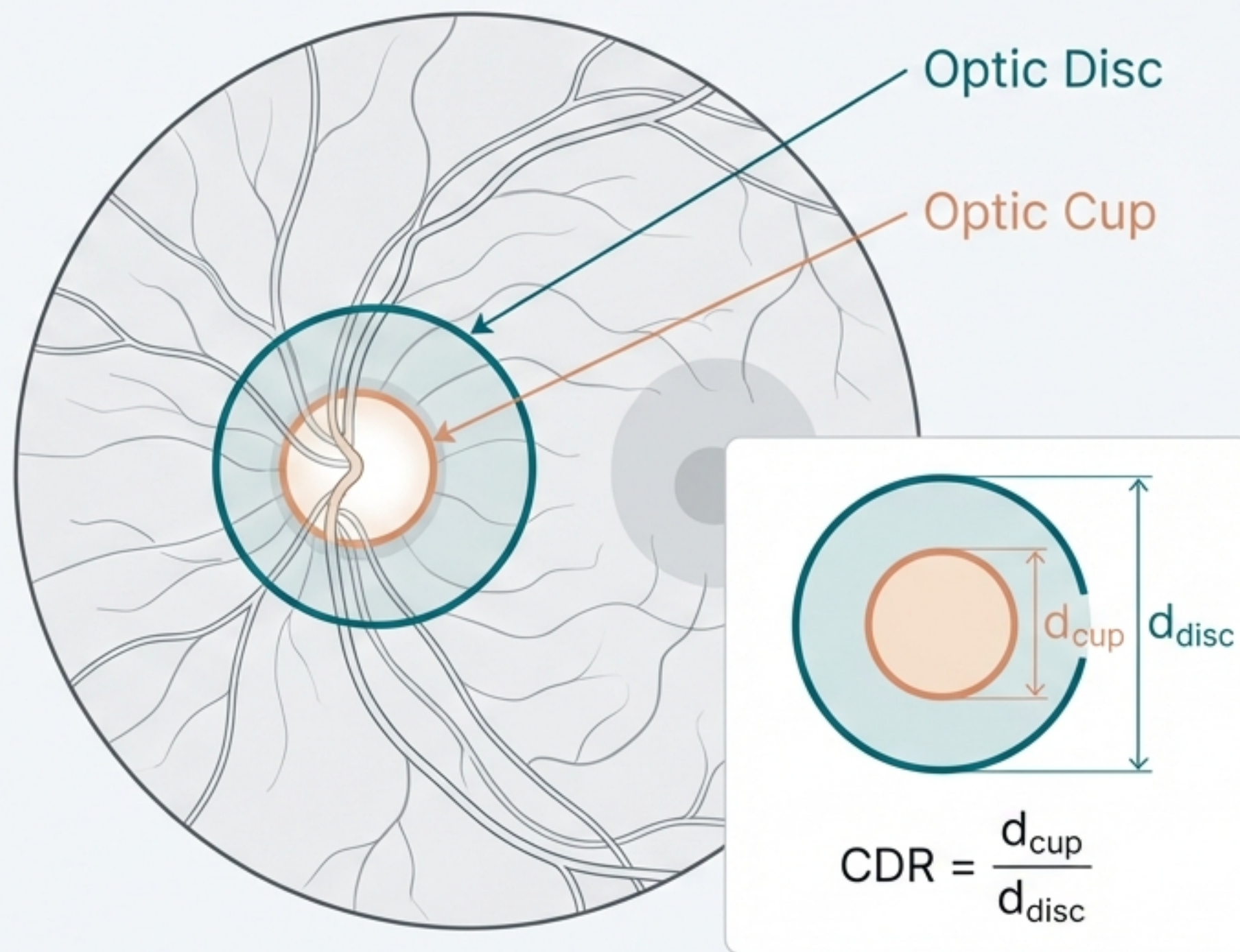
A comparative analysis of U-Net architectures and optimization strategies for robust clinical application.



The Clinical Challenge: Automating a Critical Diagnostic Marker

Glaucoma is a leading cause of irreversible blindness. Early detection is critical.

Diagnosis relies on examining the Optic Nerve Head to measure the Cup-to-Disc Ratio (CDR). A high CDR (typically > 0.6) signals disease progression.



Our Mandate: Patient Safety Overrides All Other Metrics

The primary goal is to build a high-sensitivity screening tool. Missing a diagnosis is a far worse outcome than a false alarm.



Prioritize Recall (Sensitivity):
We must minimize False Negatives to ensure sick patients are identified for follow-up.



Target: Achieve a minimum of 85% Recall to meet the safety mandate.



Implication: This system is designed as a first-pass screen, not a final diagnostic.

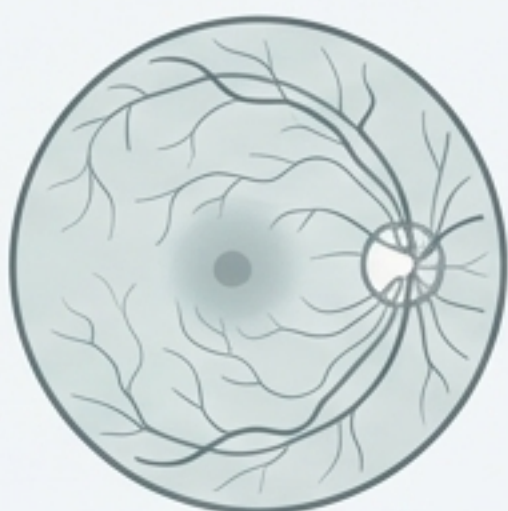
The Technical Approach: Isolating Microscopic Structures with U-Net

Core Task:

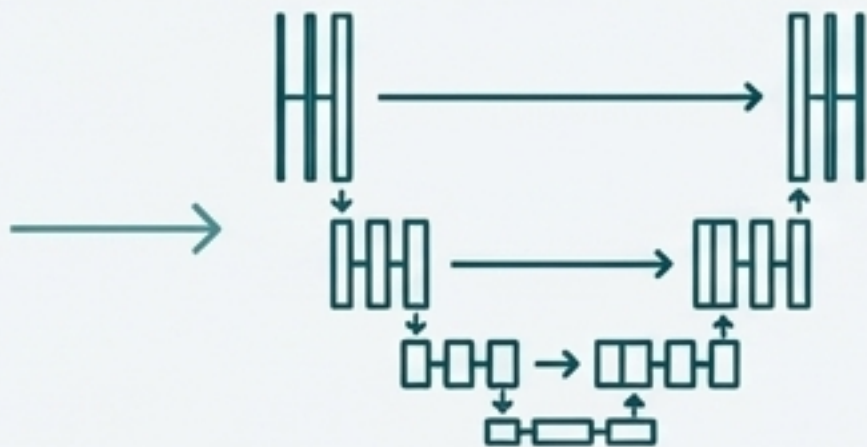
Our solution is to perform semantic segmentation of the Optic Cup and Disc. The CDR is calculated directly from these predicted masks.

The Challenge:

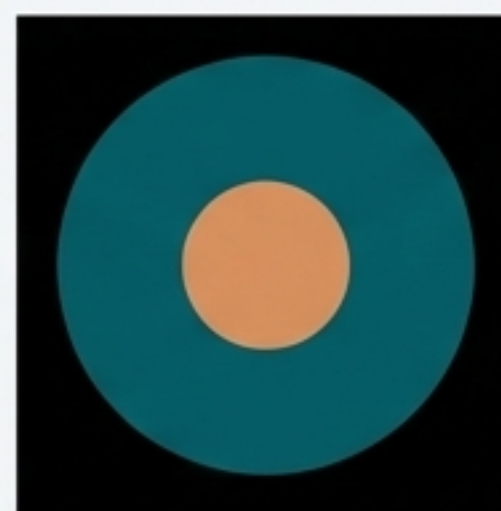
These are small, low-contrast structures with severe class imbalance, making them difficult for models to learn.



[Retinal Fundus Image]



[U-Net Model]



[Segmented Cup & Disc Mask]

CDR: 0.58

[Calculated CDR Value]

A Comparative Test of Three Architectural Hypotheses



Model 1 - The Custom Baseline

- **Architecture:** Basic U-Net
- **Parameters:** 7.8M
- **Initialization:** From Scratch

Can a simple, standard architecture solve the problem?



Model 2 - The Custom Powerhouse

- **Architecture:** Deep U-Net
- **Parameters:** 31M
- **Initialization:** From Scratch

Does a deeper, more complex custom model provide a performance edge?



Model 3 - The Expert Outsider

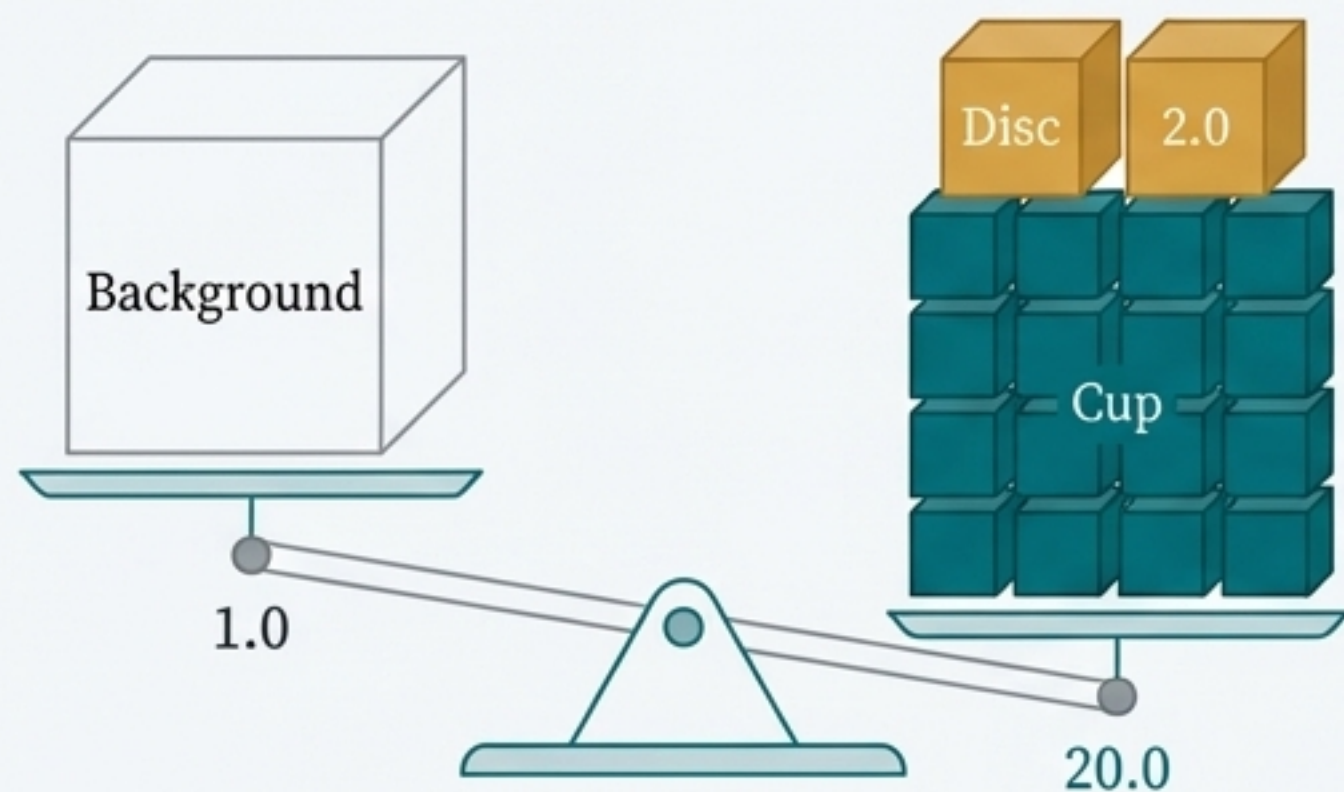
Architecture: SMP U-Net (ResNet34)
Parameters: 24M
Initialization: ImageNet Pre-trained

Can knowledge from natural images accelerate learning and improve results?

Engineering Success: A Training Strategy Tailored to the Optic Cup

Detailing the training regime for the champion Custom Deep U-Net (Model 2).

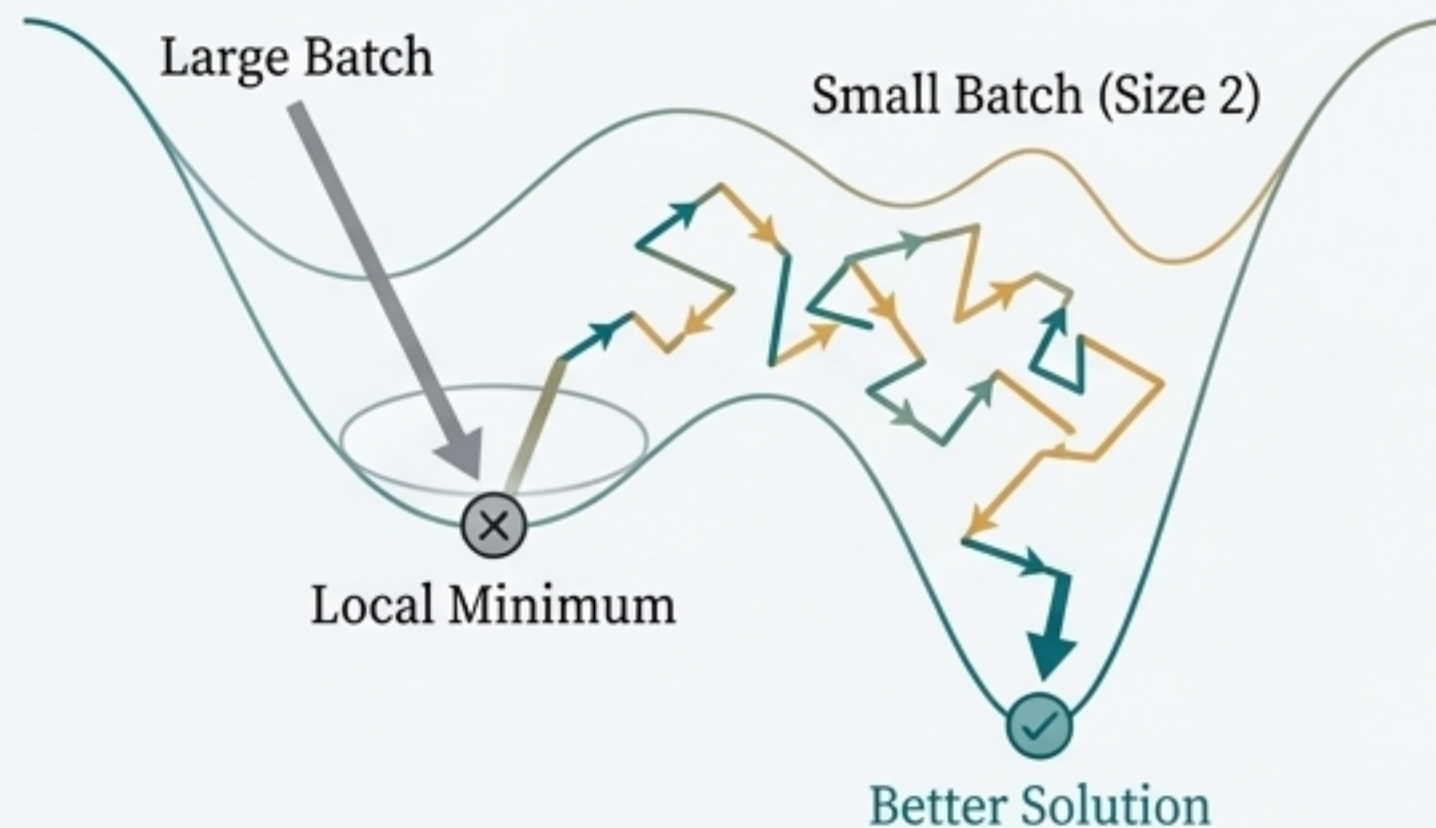
Key Tactic 1: Aggressive Loss Weighting



Detail: We used a Weighted Cross-Entropy Loss with weights of [1.0 Background, 2.0 Disc, 20.0 Cup].

Rationale: This forces the model to pay extreme attention to the tiny, clinically vital Optic Cup, overcoming the severe class imbalance.

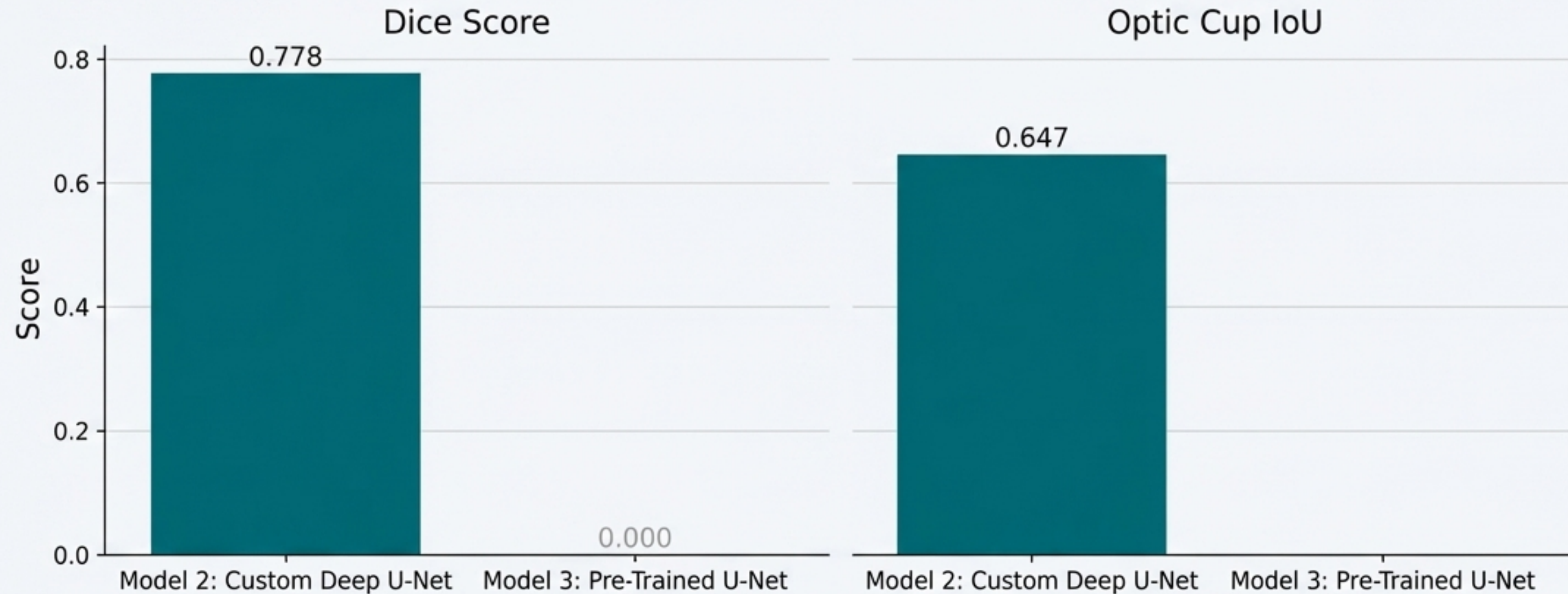
Key Tactic 2: Stochastic Optimization



Detail: Training employed a very small Batch Size of 2.

Rationale: This high stochasticity prevents the optimizer from converging on a simple local minimum (e.g., ignoring the cup entirely) and encourages exploration of the loss landscape.

Segmentation Performance: The Custom Deep U-Net Proves Superior



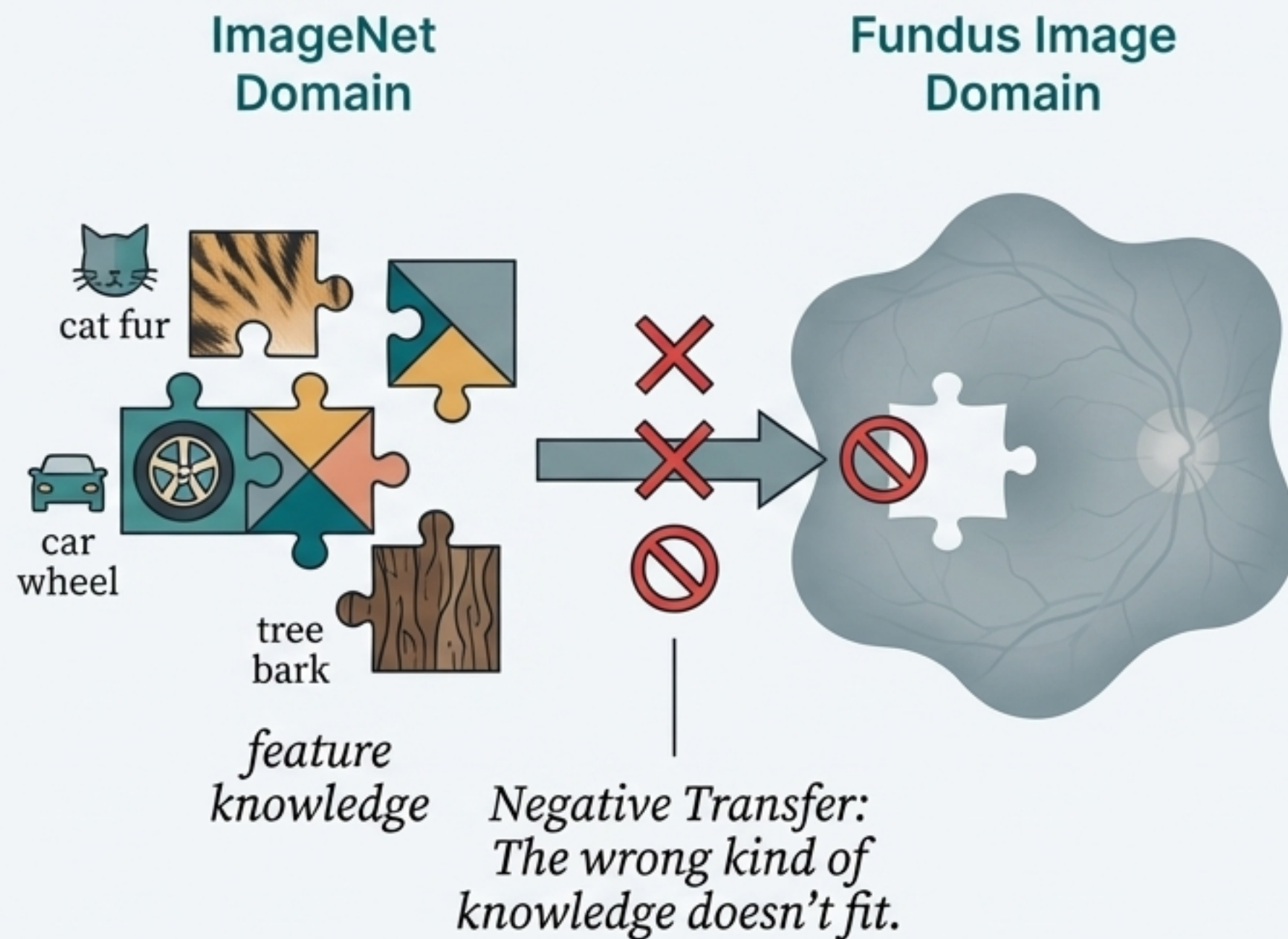
The Custom Deep U-Net (Model 2) significantly outperformed all other architectures, achieving the highest segmentation accuracy.

A Cautionary Tale: The Catastrophic Failure of Transfer Learning

Key Finding: The ImageNet pre-trained model (Model 3) completely failed to learn, resulting in a Dice Score of 0.000.

This is a classic case of **Negative Transfer** (in ochre #E29578). The features learned from natural images (e.g., edges of cars, textures of animals) are not just unhelpful—they are actively detrimental to identifying the subtle, low-contrast features of medical fundus imagery.

For this highly specialized task, building knowledge from scratch is superior to importing it from a dissimilar domain.



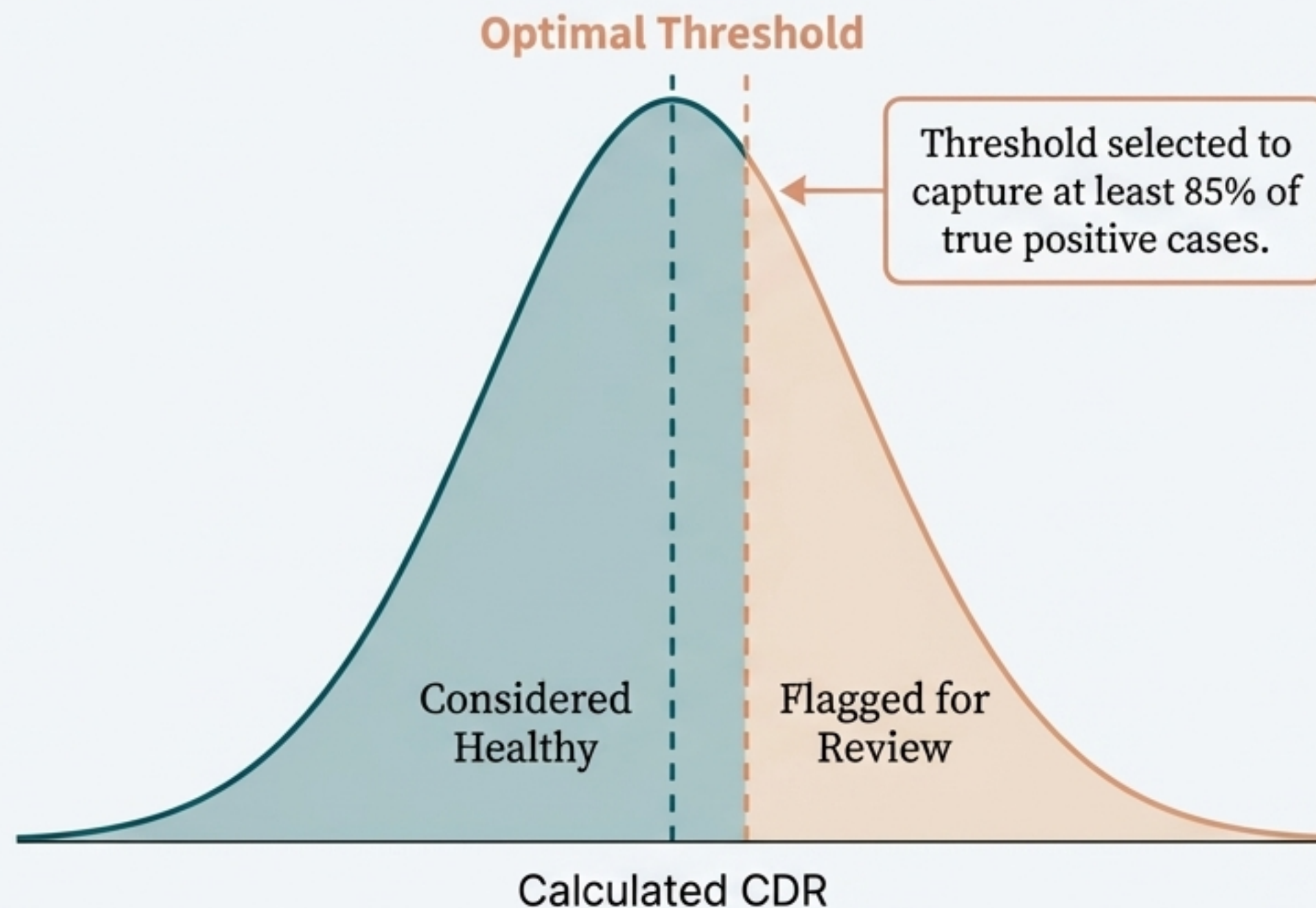
From Segmentation Masks to a Clinical Decision Threshold

Process

1. The champion model generates segmentation masks for the Cup and Disc.
2. The Cup-to-Disc Ratio (CDR) is calculated from each mask.
3. We analyze the distribution of CDR scores across our test set to find the optimal clinical cutoff.

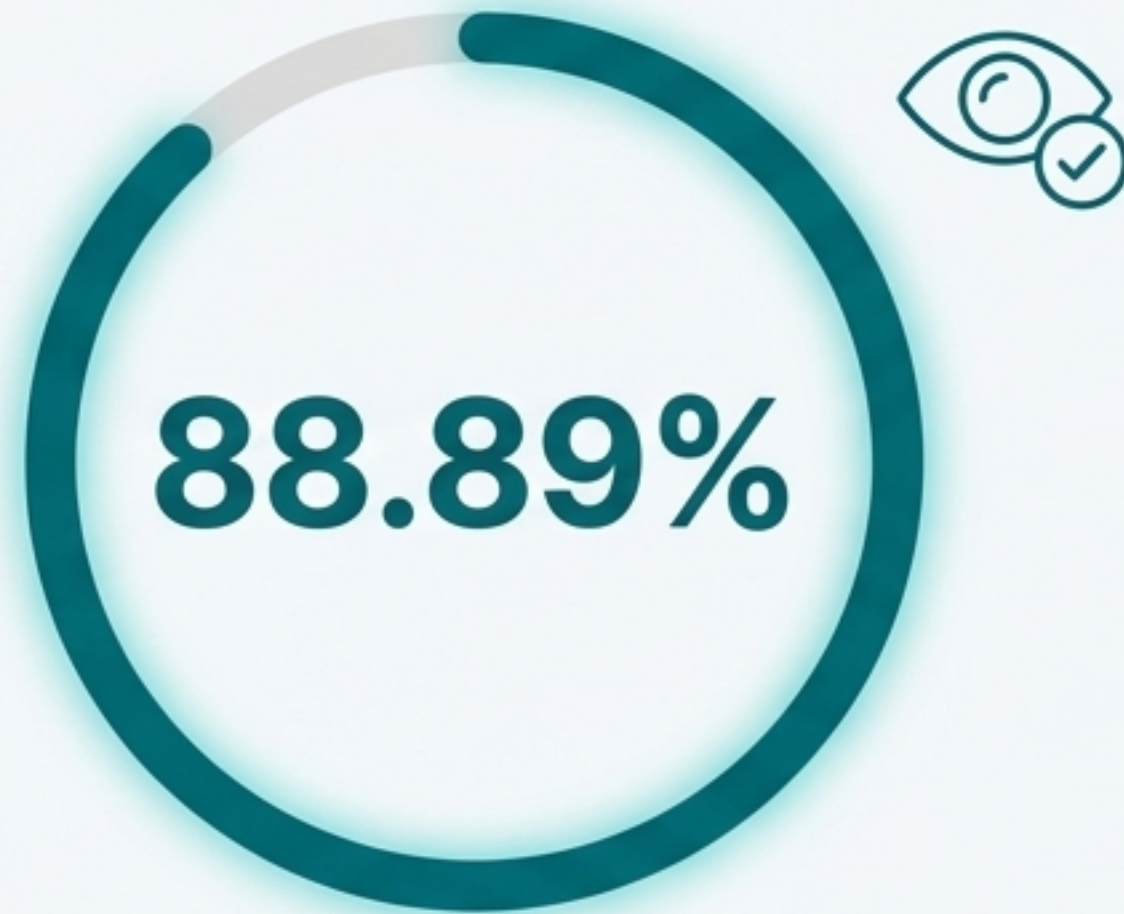
The Goal

The CDR threshold is tuned specifically to meet our primary objective: achieving a minimum patient safety (Recall) of 85%.



The Final Verdict: Mission-Critical Safety Achieved

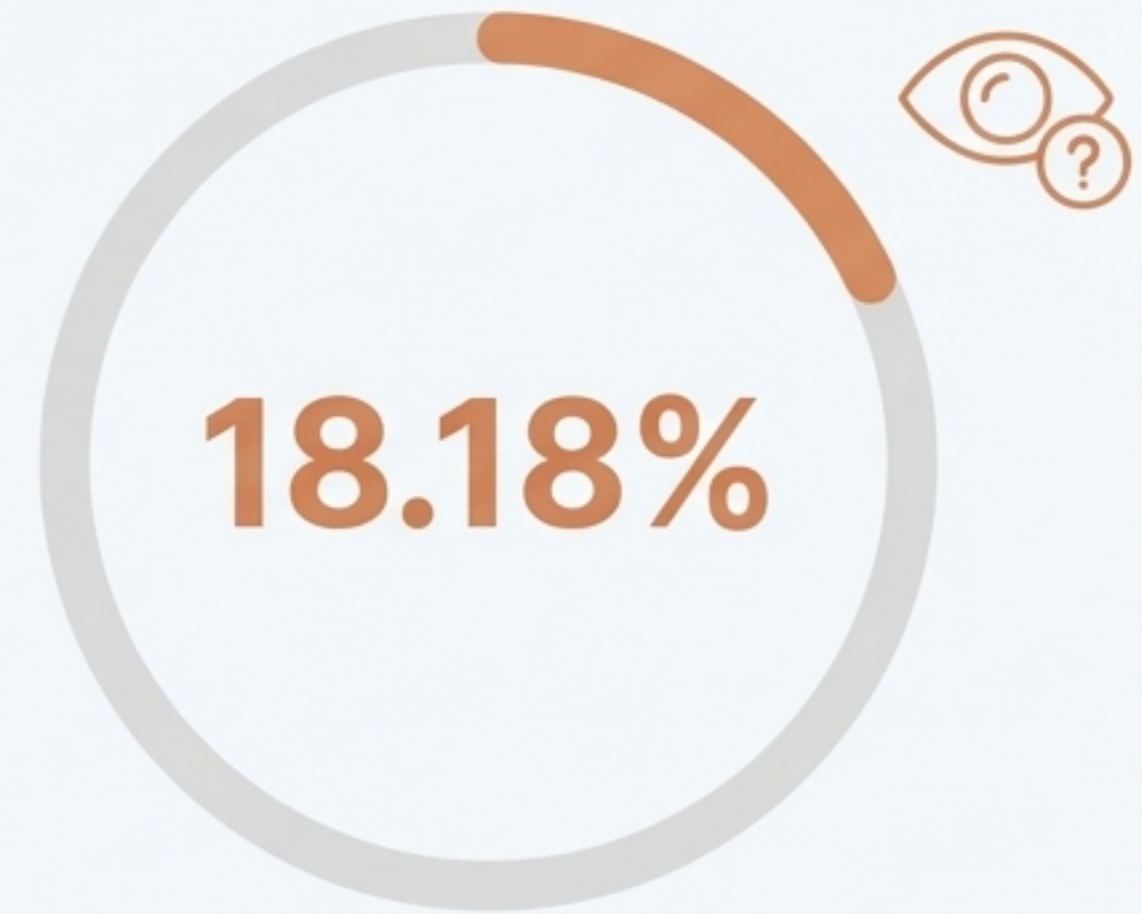
Optimal Threshold: $\text{CDR} > 0.3414$



Safety (Recall)



****SUCCESS****. Exceeds the $\geq 85\%$ clinical mandate. We correctly identify nearly 9 out of 10 patients with glaucoma.



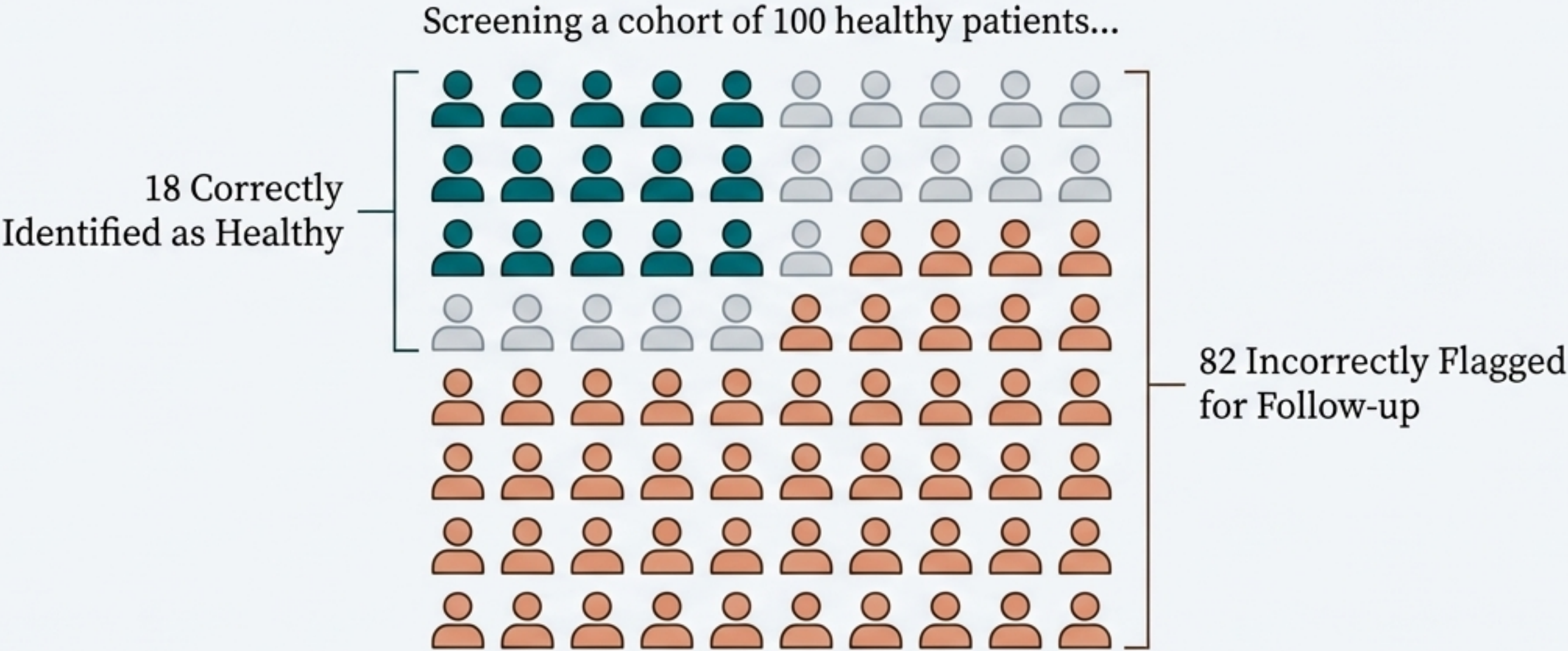
Efficiency (Specificity)



****CHALLENGE****. The system generates a high rate of false positives.

Understanding the Clinical Workflow: The High Cost of Low Specificity

What does 18.18% Specificity mean in practice?



While patient safety is guaranteed, the current system would flag approximately **82% of healthy patients**, creating a **significant burden for specialist review**. This highlights the need for **improved precision**.

Key Project Takeaways

1



Point 1: Architecture & Strategy Win: A custom-built, deep U-Net with a tailored loss function and stochastic training is the optimal solution for this specific segmentation task.

2



Point 2: Negative Transfer is a Real Risk: Pre-trained weights from disparate domains like ImageNet can be actively harmful in specialized medical imaging, and training from scratch should be the default approach.

3

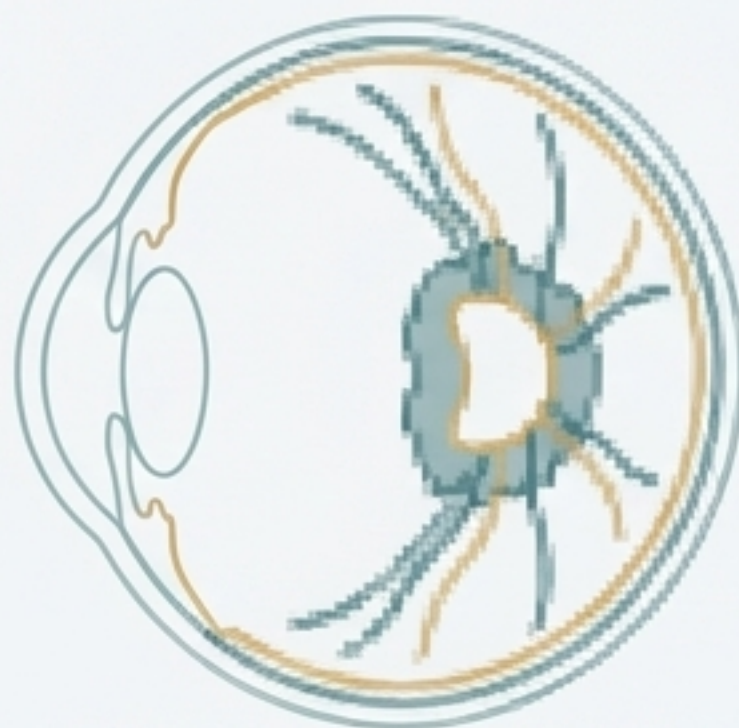


Point 3: High-Sensitivity Goal Met: The system successfully functions as a high-recall screening tool, meeting its core clinical mandate for patient safety with a sensitivity of 88.9%.

The Next Horizon: Improving Precision with Hybrid Loss

The Core Limitation

The model's primary weakness is low specificity, driven by imprecise boundary segmentation.



Current: Imprecise Boundaries

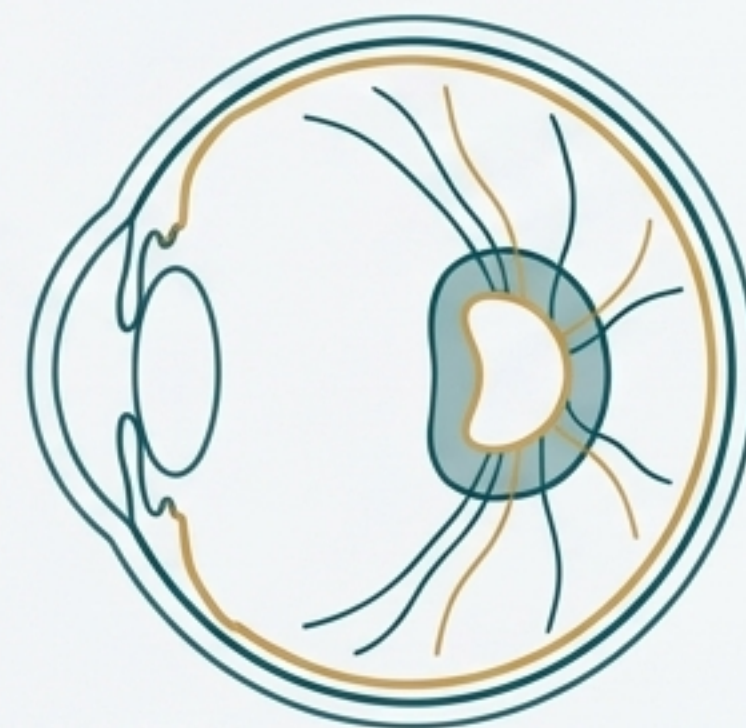


The Actionable Plan

Implement a Hybrid Loss Function: Combine the benefits of Weighted Cross-Entropy (for class imbalance) with a Dice Loss component (for improving boundary overlap).

The Goal: Push the segmentation Dice Score over 0.80.

The Expected Outcome: A significant increase in Specificity while maintaining the high Recall already achieved, making the tool clinically viable and efficient.



Future: High-Precision Boundaries